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Bullet Hole Detection in a Military Domain Using Mask R-CNN and ResNet-50

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***Abstract*—Small arms shooting practices and competitions are routine activities in the military domain. The shooting group or bullet group analysis serves as a metric for the precision of a weapon, the shooter's accuracy, and consistency, and as a method for improving or refining one's shooting abilities. This analysis mechanism, however, is either manual or semi-automatic, employing image processing-based algorithms such as template matching, histogram equalization, white balancing, median and gaussian altering, peak detection, and image subtraction in an indoor setting, which is incapable of adapting to environmental conditions such as humidity, temperature, ambient light, wind speed, and rain, among others. Recent advancements in artificial intelligence or deep learning techniques explored ways to facilitate automation in various sectors. In this paper, we have used such deep learning approaches to automize the shooting system in real-time within a military domain and achieved success in resolving the traditional image processing drawbacks. Our proposed methodology has two phases. The first phase uses Mask R-CNN a conceptually simple, flexible, and general framework for object instance segmentation to extract the target region from the environment, and in the second phase, we fed the output segmented target of the first phase to ResNet-50 a convolutional neural network architecture to detect the bullet holes. Several experiments have been conducted on real-time datasets and the results show 0.87 of average precision using mask R-CNN to segment the target and ResNet-50 give 0.80 to detect bullet holes.**

***Index Terms*—deep learning, machine learning, computer vision, bullet hole detection**

I. Introduction

The military generally evaluates the efficacy of shooting using a semi- or fully-manual approach. Figure 1 shows shots fired from different places and positions in multiple phases to determine whether the bullets hit the target inside a 10-inch diameter. If yes, he gets credit; otherwise, a washout. A ruler measures this 10-inch diameter. After shots are fired, the observer approaches the target, analyzes the impact using the ruler, and records the data. Entry of the conclusion is done manually into a database or records. This strategy is neither quick nor efficient and demands too much labour. Therefore, the existing system of shooting scores must be revised, and an automated firing assessment system is required.

People have been motivated, in part by technological progress, to automate and modernize the scoring system for shooting competitions. The automatic tally of shooting scores has been accomplished using a wide variety of image processing methods. These include perspective transformation for scanning and cropping the distorted image, image subtraction, erosion, dilation, size normalization, template matching, Hough transformation, image binarization, geometric rectification, edge detection, shape detection, thresholding, and filtering. [1], [2], [5] [15]–[18]Although several methods have been offered for analyzing target shooting scores, most of these approaches have been developed for use in the sporting arena rather than the military. The results can be heavily influenced by environmental factors that are ignored by the proposed automatic fire-scoring methodologies in the literature.

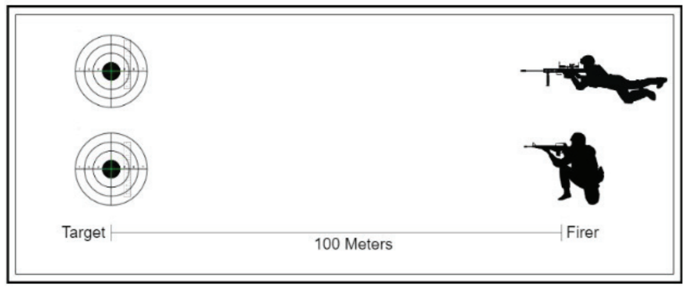


Fig. 1. Army firing System

Several machine learning algorithms for the processing of pictures have reached great precision for a variety of applications, such as for segmentation, in recent years. Mask R-CNN, which has achieved 92.68 percent recall rate and a 97.58 percent accuracy [19] respectively, is one method for the detection and localization of objects. Mask R-CNN might therefore be utilized to solve the issue. In order to adapt to the external environment and attain high precision, machine learning algorithms always obtained successful outcomes.

Based on our military domain, in this paper, we proposed two algorithms in succession to achieve our objective.

- The Mask R-CNN for target extraction from the environment since it is used to address instance segmentation issues in computer vision, which is crucial when attempting to recognize distinct objects inside the same picture by recognizing the object's bounding box and classes. [13]
- ResNet-50 was utilized for bullet hole identification since it is a pre-trained deep neural network that can be mod-

ified for various use cases. We know that training with 8,000 pictures enables the CNN to attain 77% accuracy, whereas ResNet-50 delivers 98% accuracy, based on work observed in other kernels. [14]

The paper will initially present the main points of a concise literature review. Workflows will be used to illustrate the methods of this research in depth. The discussion will conclude with experimental results and final remarks.

II. Literature Review

The relevant literature is reviewed under categories including a software framework, image processing methods, bullet hole detecting approaches, and machine learning algorithms.

A. Image Processing

In [1] authors aim to automatically identify bullets and provide grading that complies with Army Standard Operating Procedures (SOP). Only indoor firing can use bullet detection with any degree of accuracy. Due to the complex target structure and outside lighting circumstances, it is challenging to see bullet holes. When it is found that an error has been made, a correction is put into place. However, there was no discussion of the soldier's performance's statistical forecast. [1] Another study describes an automatic scoring system for live shooting sessions based on image processing where a camera is installed in front of the shooting target frame to record each individual shot. It employs a variety of image processing algorithms, including target ring identification, perspective transform, image subtraction, and morphological image processing. They employed 10 target sheets, each of which contained 10 bullet hole photos recorded using a circular sticker. Utilizing computer vision to detect targets is far less expensive than using a shooting target scanner or an optical evaluation system, according to this study that employed image processing techniques. The automatic shooting scoring architecture consists of four main processes which are image perspective transformation, scoring circle detection, bullet hole detection, and scoring mechanism. The image subtraction method is a good method to detect bullet holes from each shot inside the target sheet, but in case of the sensitivity of the image subtraction method, it is easy to make noises because of the change of external conditions in example light and other unwanted events, can even influence the image subtraction process. Some noises occurred in the image subtraction process because of the sensitivity of the image subtraction method and make the scoring process not precise. [5]

Using image processing in a separate study, researchers were able to eliminate the influence that gunshot holes had on the background. The treatment is extraordinarily expensive and time-consuming. This was designed particularly for indoor shooting. A bullet may be detected with a 100% degree of accuracy. Researchers applied background subtraction and Euclidean Distance Methods in building a shooting scoring system automatically. Background Subtraction Technique functioned to detect the former shot holes and Euclidean Distance to measure the distance between holes shot against the centre point and at the same time enabled for automatic score calculation. This research produces a system that is able to run well under conditions of normal lighting. The results of the shot scores can be displayed on the monitor on the side of the gunners. Image processing is carried out in the below steps:

1. The process of transforming the gunshot image 2. The process of cropping the shot 3. The process of detecting circle assessment 4. The process of detecting shot marks 5. Automatic assessment process.

The intensity of light affects the detection results of the shot hole. In very dark lighting conditions light will reduce the level of detection accuracy. Normal lighting conditions are measured in the range of 80 lux s.d. 120 lux is able to detect properly and level of accuracy in detecting under normal conditions until it reaches 100

B. Deep Learning in bullet hole detection

Based on surveillance recordings of the target, another paper attempts to localize the bullet holes on a 4m*4m target surface and identify the shot time and position of fresh bullet holes on the target surface. Using two networks in sequence, an enhanced model based on Faster-RCNN is proposed to handle the problem in this research. The first network is trained using original video frames to acquire coarse bullet hole locations, while the second network is trained using the candidate sites obtained by the first network to obtain precise locations. On our bullet-hole dataset, the series Faster-RCNN algorithm improves the average precision by 20.3% compared to the original Faster-RCNN technique. There was only one type of target in the bullet hole detection task, and each image contained 5 to 10 bullet holes, so the size of the dataset was basically enough. After all the frames had been processed, it could obtain a list of all bullet shots on the target. In this paper, a concatenate network structure based on Faster-RCNN is proposed. Frames from target surveillance video pass through the coarse localization network first, and then input to fine localization network, and finally, output precise locations of bullet holes on the target. The network achieves better performance in the bullet database compared to the original Faster-RCNN model, and it provides some new ideas for solving the problem of small object detection. However, the proposed method only performs well on small datasets. TensorFlow which is used here is a platform that can be utilized for object detection in computer vision. This study included a comparison of R-CNN, Fast R-CNN, and Faster R-CNN. Faster R-CNN is proven to have better accuracy while requiring a sophisticated computation procedure. [3]

C. Research Gap

As mentioned earlier, the previous methods were incapable of working in real-time with surroundings located outdoors. The hardware configuration that was proposed was not enough for use in field tests. Additionally, The strategy of image subtraction did not work that well because of weather and camera adjustments. As a result, the before and after images taken on the ground will never be an exact match pixel for pixel. Moreover, none of the methods has to address the military arm shooting problem using deep learning methods.

III. Methodology

We divided the whole methodology part into two phases:

- Phase 1: Use of Mask R-CNN for target detection and segmentation
- Phase 2: Use of Mask ResNet-50 for bullet hole detection

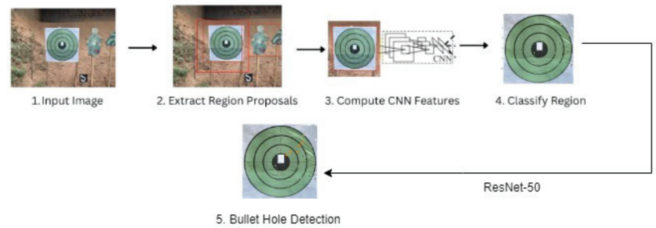


Fig. 3. Graphical Proposed Methodology

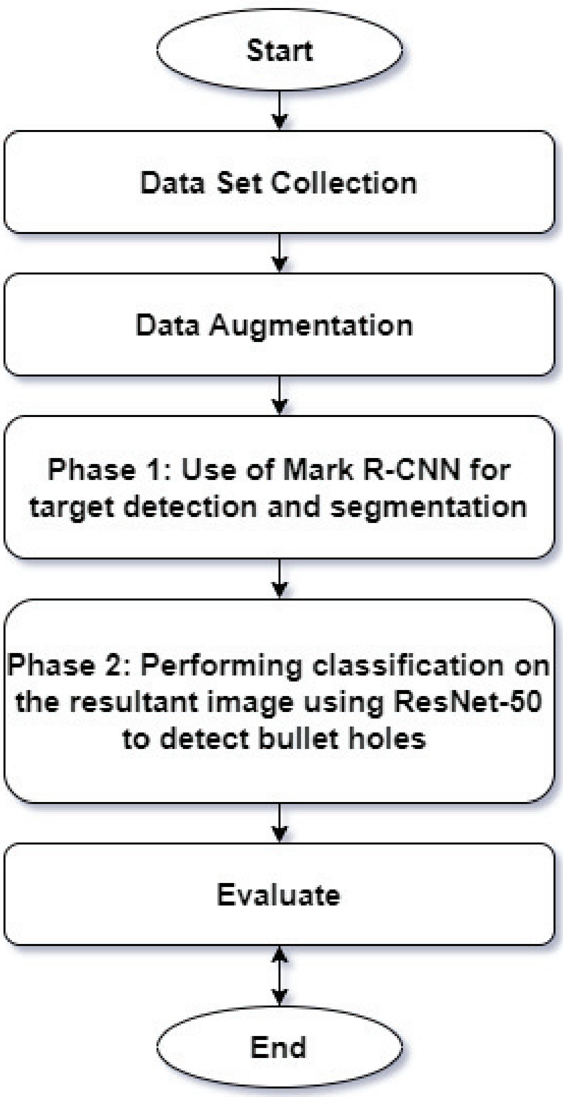


Fig. 2. Proposed Methodology

In Figure 2 we have demonstrated the whole methodology. Each component is executed in the order illustrated in Figure 2, with the result of one component acting as the input for the next. This suggests that the interaction between components consists of a succession of consecutive actions. Initially, image data was acquired for the target using various settings. Then, our training data were augmented, which increased our accuracy. Following the segmentation of the target region employing Mask R-CNN, ResNet-50 recognized the gunshot holes. After the ResNet-50 findings were computed, the fire was assessed. Figure 3 is a demonstration of the overall architecture working behind our system.

A. Data Set Collection

The data set was manually collected in the form of photographs taken from various angles using a camera. The primary objective was to offer as much environmental diversity as possible. Such as sunshine, shadow, wind, rain, and low light. The photographs were taken at various times of day and from various distances from the target. The collection of 400 images was followed by preprocessing and augmentation. Figure 4 illustrates samples of our collected data. We labelled the gathered images using CVAT, an Intel computer vision annotation tool, to create the custom dataset for object detection. We required two distinct datasets. The first is for Mask R-CNN with the target region labelled, while the second is for ResNet-50 with bullet holes labelled.

Fig. 4. Collected Data

B. Data Augmentation

During the training phase, the cropping, padding, rotation, translation, affine, brightness, contrast, and saturation value settings were all randomized. This allowed for the generation of enhanced pictures.

Convolutional neural network architecture performs effectively with a huge dataset that includes a diverse set of data points. [9], [11] Adding slightly modified copies of already existing data or producing new synthetic data from previously available data are examples of data augmentation approaches. It acts as a regularizer to help minimize model overfitting during training. It can accommodate both the variety and volume of data used for training. Besides these two uses, enhanced data can be used to address the problem of class imbalance in classification tasks. Rescaling, rotation (within a 360° range), horizontal flip, vertical flip, etc. are some of the factors used for data augmentation. The time-consuming computations are sped up by transforming all the data photos into NumPy arrays.

C. Phase 1: Use of Mask R-CNN for target detection and segmentation

Mask Regional Convolutional Neural Network (Mask R-CNN) is a powerful object detection algorithm based on Faster R-CNN, which complements and extends its architecture to incorporate an additional layer of segmentation for each instance. [8] Mask R-CNN provides a third output, termed Mask, in addition to the labelled class and bounding box for each discovered item. Parallel to the location and identification, this deep neural network-based method is able to generate the recommended locations where the objects might be discovered by segmenting instances for each bounding box using a fully connected layer (FC). [10] From a high enough vantage point, Mask R-CNN can be broken down into its three component parts: the backbone, a convolutional neural network (typically ResNet, VGG, or ConvNet) with the function of getting the characteristics of the analyzed image from a low level, such as edges, up to a high level, such as people, animals, among others. [9] The second is a light neural network called RPN (Region Proposal Network) that is responsible for receiving the map of features generated by the backbone in order to locate the regions of interest (RoI), therefore, the areas that can contain the objects. In the third scenario, a fully convolutional network (FCN) is applied to the RPN-suggested regions in order to produce three outputs for each region of interest (RoI): a classification of the object within the RoI, a bounding box with the goal of refining the location and size of the structuring element, and a mask. Bounding box, classification, and segmentation mask scores are all calculated in the last case. Figure 5 depicts the overarching structure of the Mask R-CNN.

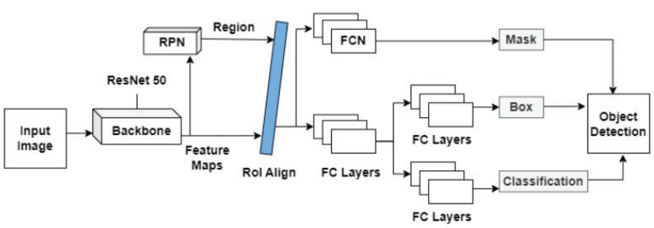


Fig. 5. Mask R-CNN architecture

We must not only detect, but also extract the target region from the environment, which is the green square. Detection of the target zone is therefore one of the primary components that will enable the identification of impacts on the firing target. The process flow begins with the receipt of pre-processed images, the loading of the trained model, the execution of the model for detection, and the segmentation of the selected target from its surrounding environment. As described Mask R-CNN has devised a simple classifier to predict whether or not the to-be-identified area falls within the silhouette. Then, it returns three suggestions: the categorization (class), the bounding box of the bullet hole (box), and the mask of pixels of the target region (mask). The trained model Mask R-CNN is loaded with its preset specifications, which include the backbone network (based on ResNet), batch size, learning rate, mask shape, and other configurations listed in table 1.

Figure 6 depicts the use of Mask R-CNN in our project to extract the target region from the environment. This process utilizes pixel-to-pixel alignment and bounding boxes to categorize several image areas into the suggested class by producing a segmentation mask.

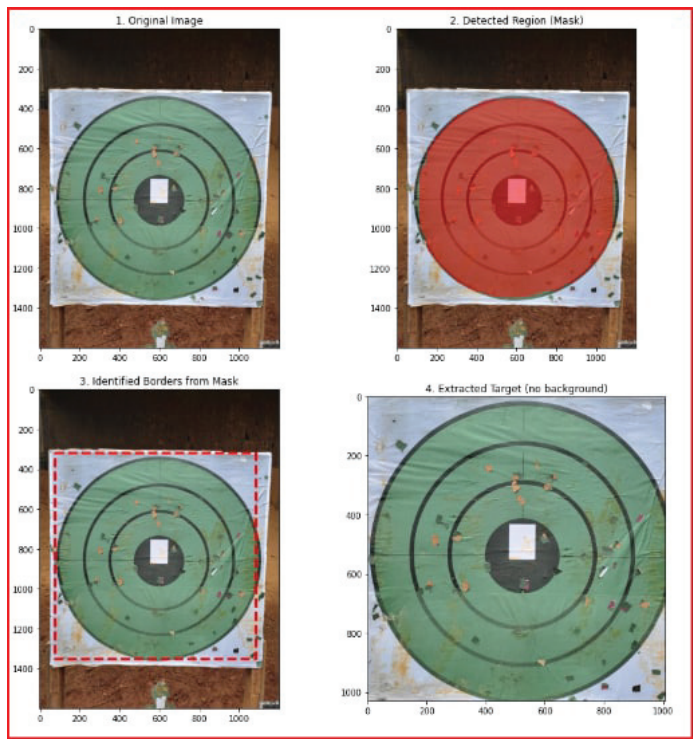


Fig. 6. Mask R-CNN in Segmenting the target area

D. Phase 2 : Use of Mark ResNet-50 for bullet hole detection

Deep convolutional Residual Networks use shortcut connections to skip blocks of convolutional layers. The "bottleneck" blocks obey two simple design rules: (I) for the same output feature map size, the layers have the same number of filters; and (ii) if the feature map size is half, the number of filters doubles. Convolutional layers with a stride of 2 immediately down-sample and batch normalize after each convolution before ReLU activation. The identity shortcut is used for

TABLE I

Parameters	Epochs	Steps per epoch	Optimizer	Learning rate	Learning momentum	Weight decay	Batch size	Backbone strides	Mask Shape
values	80	50	SGD	.001	.9	.0001	2	[4,8,16,32,64]	28*28

TABLE II

Name	Batch Size	No of epoch	No of warm-up epoch	Learning rate base	Warmup learning rate base	Decay drop	Step size	No of warmup steps 1st	Total steps 1st
Value	16	15	2	1e-4*2	1e-3*2	0.5	Batch Size	2*Step Size	No of war-up epoch* Step Size

identical input and output dimensions. The projection shortcut matches dimensions with 1×1 convolution as dimensions rise. Both shortcuts cross feature maps of two sizes with a stride of 2. A 1,000-FC layer with softmax activation concludes the network. 50 weighted layers have 23,534,592 trainable parameters. Figure 7 shows the original ResNet-50 design. [11]

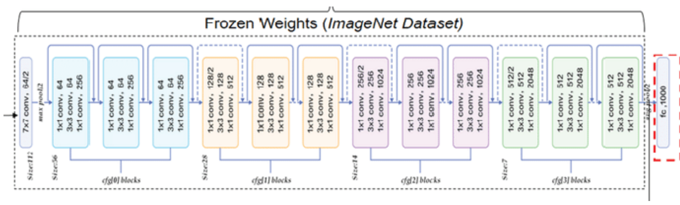


Fig. 7. ResNet-50 Architecture

ResNet50 is a pre-trained Deep learning model. Since ResNet-50 has excellent image classification performance and can extract high-quality image features also the accuracy and depth can be improved without difficulty, we hypothesized that its feature extraction layers would also perform well in detection. Therefore, we trained ResNet-50 using ImageNet to identify the bullet holes. This model's input was the image produced by Mask R-CNN. The image was fed into the ResNet50 layer with the pre-trained weights, and the final layer was a traditional fully-connected dense layer with softmax activation as seen in Figure 8. Importing Pre-Trained Weights for the ResNet50 model, the input data was trained using pre-learned weights, and only the dense layer was learned through backpropagation. A few layers, such as Batch Normalization (BN) layers, were not frozen since the dataset's mean and variance were unlikely to match those of the pre-trained weights. Therefore, auto-tuning was modified for the BN layers in ResNet50, so that a few of the layers at the top of ResNet50 were not frozen. Batch size, no of epochs and other parameters used during training are mentioned in Table 2. Figure 9 is the final output of the overall system.

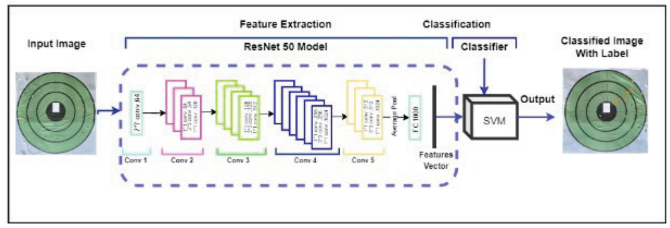


Fig. 8. ResNet-50 in detecting bullet holes

Fig. 9. ResNet-50 Final Output

IV. Result Analysis

A. Mask R-CNN

Figure: 10 depicts the confusion matrices obtained for test data at the completion of training.

Four hundred pictures were created from the mask RCNN, of which 350 were chosen for the train data and 50 were chosen for the test data. On the training dataset, the accuracy of the model was 1 as the images were almost similar, however, on the validation dataset, it showed a mean average precision of 0.83. (see Figure 11).

B. ResNet-50

In tensorboard, loss functions depict how the model optimizes to generalize better. Since the optimization process minimizes the global loss, the regularization term (which is

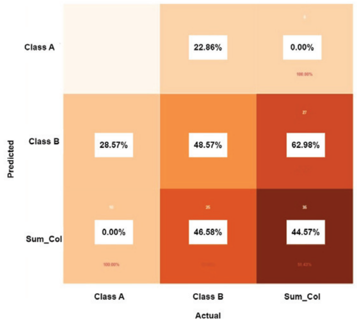
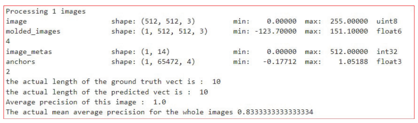


Fig. 10. Confusion Matrix



```
Processing 1 images
image                  shape: (512, 512, 3)         min:    0.00000  max:  255.00000  uint8
molded_images          shape: (1, 512, 512, 3)      min: -123.70000  max:  151.10000  float6
4
image_metas            shape: (1, 14)               min:    0.00000  max:  512.00000  int32
anchors                shape: (1, 65472, 4)         min:   -0.17712  max:    1.05188  float3
2
the actual length of the ground truth vect is :  10
the actual length of the predicted vect is :  10
Average precision of this image :  1.0
The actual mean average precision for the whole images 0.8333333333333334
```

Fig. 11. Mask RCNN Precision and Accuracy

large when the weights are far from zero) will steer the optimization toward solutions with close-to-zero weights. The regularization loss function decreases gradually in the case of ResNet-50 shown in Figure 12. Epoch being set to 15 where with the change of weights the loss function tends to decrease which shows the improvement of the model in detecting bullet holes. Training accuracy of the model was 89.2% whereas a validation dataset with 30 separate images produced a detection accuracy of 86% for the bullet holes. Training loss and validation loss were .1124 and .1409 respectively.

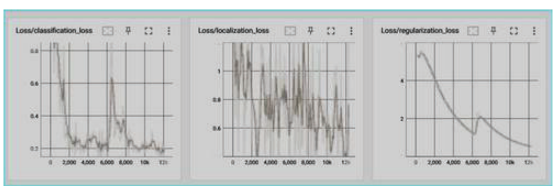


Fig. 12. Loss function

V. Conclusion

For this study, we employed Mask R-CNN for target identification and segmentation, whereas ResNet-50 was utilized to identify bullet holes. Cropping, padding, rotation, translation and brightness changes were used to augment the datasets after they were gathered from actual ground photos and then used in the collection process. Mask RCNN's mean average precision for cropping the initial image was 83.33%. In the case of ResNet-50, the regularization loss decreased gradually to detect the bullet holes and ultimately gave an accuracy of 86% in the validation dataset. Even though the results of both phases are satisfactory while tested using real-time images, the accuracy decreases as a consequence of the images' declining quality and an increase in the number of repairs made to cover up earlier bullet holes. In spite of that, we are able to draw the conclusion that a system similar to this one will be of great use to us when it comes to training and exercises in the military sectors.

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